

Phase D5 Key Prior-Art Claims Index

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Phase D5 of the CKS derivation-note series establishes operational tests for Paper 3’s governance commitments. This note (#649) serves as the key claims index for Phase D5, collecting the ten prior-art claims that the 60-test library and its supporting notes establish, organized for navigation and future citation. Claim D5-01 is the meta-claim: Paper 3’s governance commitments are operationally precise enough to be verified from governance records alone, foreclosing the adversarial argument that prior art in this space is merely vague aspiration. Claims D5-02 through D5-10 are specific instances of that general testability property, each covering a distinct dimension of Paper 3’s architecture. Claim D5-10 adds a self-assessment property — the validation honesty of the test library itself — that strengthens the corpus’s credibility as a genuine prior-art contribution. Phase D5 closes with note D5.15.

1. Purpose and position of this note

Phase D5 of the CKS derivation-note series (Series D, Paper 3 derivation) comprises notes D5.01 through D5.15. The phase builds a 60-test library for Paper 3’s governance commitments and develops the supporting framework for using those tests in practice. Notes D5.01 through D5.13 are complete.

This note, D5.14 (#649), is the key claims index for Phase D5. Its purpose is navigation: to collect, name, and briefly describe the ten most significant prior-art claims that Phase D5 establishes, together with the notes that establish each claim, so that future work in Series D, Series CC, and Series T can cite Phase D5 claims with precision.

Phase D5 closes with note D5.15, which provides final closure for the phase and transitions to Phase D6 (boundary cases).

2. Phase D5’s affirmative contribution to the prior-art corpus

Phase D5 occupies a specific position in the derivation-note series. Phase D2 established what the governance requirements of Paper 3’s architecture are — the operational variants and sub-commitments that

follow from Paper 3's six named claims. Phases D3 and D4 established failure modes and composition pairs. Phase D5 establishes something different: that the requirements already in the record are testable and measurable.

This distinction matters for the prior-art record in two ways. First, a prior-art corpus of vague commitments is vulnerable to the adversarial argument that it names aspirations rather than architectural properties. Binary testability forecloses that argument: if a governance commitment can be verified or falsified from governance records alone, it is not vague. Second, measurability extends the prior-art coverage from static architecture into governance-quality monitoring — an active, ongoing dimension of any deployed system. Phase D5 claims span both of these contributions.

3. The ten key prior-art claims

Claim D5-01: Operational precision of Paper 3's governance commitments

Statement:

The 60-test library establishes that Paper 3's governance commitments are operationally precise enough to be binary-tested from governance records alone, foreclosing the "vague prior art" adversarial argument.

Why this is the meta-claim:

D5-01 is both the logical foundation of Phase D5 and the summary of its contribution. If this claim were false — if Paper 3's commitments were too vague to binary-test — Claims D5-02 through D5-10 would be incoherent. The existence of 60 tests that satisfy three validation criteria (binary from records, distinct coverage, non-trivial pass/fail) is the proof that the claim holds. Every more specific claim that follows is an instance of this general property.

Established by: D5.07 (complete 60-test library), D5.13 (Phase D5 synthesis).

Claim D5-02: Pre-authorization as the binary-testable solution to the speed-governance tension

Statement:

Pre-authorization is the binary-testable solution to the speed-governance tension — time-sensitive full-aspect integration events can be governed by verifying that pre-authorized standing configurations were used, not by requiring real-time governance authoring.

Significance:

The speed-governance tension is a structurally recurring objection to any governance framework applied to time-sensitive coordination: if governance requires human authoring at event time, it will be bypassed under time pressure. Pre-authorization resolves the tension architecturally. Binary testability is what makes the resolution usable as prior art: an assessor can verify from governance records whether standing configurations were pre-authorized and whether the event used them.

Established by: D5.02 (Test 21).

Claim D5-03: Binary testability of N-ary governance completeness

Statement:

N-ary governance completeness is binary-testable — for N greater than two participants in a full-aspect integration event, the presence of three-way joint authorization and cross-pair conflict routing can be

verified from governance records.

Significance:

Paper 3's governance architecture is specified for the two-party case but extends to N participants. This claim establishes that the extension is not merely theoretical: the presence of joint authorization and conflict-routing completeness for all pairs is verifiable. This forecloses a narrow reading that limits the prior-art coverage to two-participant scenarios.

Established by: D5.02 (Test 22).

Claim D5-04: Binary testability of the competition variant with preserve-tier routing

Statement:

The competition variant combined with preserve-tier routing is binary-testable — the presence of both competition variant selection and preserve-tier routing rules can be verified from governance records, detecting misrouting of preserve-tier content.

Significance:

Paper 3 defines three full-aspect integration pattern variants (collaborative, competitive, complementary) and a three-tier conflict-handling mechanism (preserve, resolve, escalate). The interaction between the competition variant and the preserve tier is a specific failure-adjacent configuration requiring explicit governance attention. Binary testability of both together provides targeted detection of the anti-pattern in which preserve-tier content is misrouted under competitive selection conditions.

Established by: D5.02 (Test 23).

Claim D5-05: Governance quality is quantitatively measurable

Statement:

Governance quality is measurable through test pass rates, trackable over time, and integrable with health monitoring — governance quality has a quantitative dimension.

Significance:

Most governance frameworks produce binary assessments (compliant or non-compliant at a point in time). This claim extends the prior art into the quantitative dimension: governance quality can be expressed as a pass rate across the test library, tracked as a time series, and integrated with health indicators to create a governance quality assurance cycle. This dimension of the prior-art coverage is novel relative to the binary-compliance framing that dominates governance literature.

Established by: D5.08 (calibration guidance), D5.12 (health integration).

Claim D5-06: Six scenario-specific test subsets for practical deployment

Statement:

Six scenario-specific test subsets make the 60-test library practically usable across different governance contexts without requiring all 60 tests for every event.

Significance:

A 60-test library applied uniformly to every governance event would be operationally impractical in most deployment contexts. The six scenario-specific subsets — organized around governance audit, event readiness, population assessment, evolution tracking, conflict assessment, and cross-paper compliance —

make the library selectively applicable. This claim is prior art for the pattern of subset-structured test libraries, distinct from a claim about any individual test.

Established by: D5.09 (scenario-specific subsets).

Claim D5-07: Binary testability of cross-paper lifecycle and evolution isolation

Statement:

Cross-paper lifecycle and evolution isolation are binary-testable — the version-fixing principle established in earlier derivation notes produces verifiable records distinguishing shared-substrate versions from home-governance-updated versions.

Significance:

Paper 3 inherits from Papers 1 and 2, and Paper 3's commitments interact with commitments from prior phases. The version-fixing principle — that a participating party's shared-substrate version is fixed at participation time and does not propagate home-governance evolution — is a cross-paper architectural claim. This claim establishes that the isolation it describes is verifiable, closing a gap where cross-paper claims might otherwise be viewed as unverifiable assertions.

Established by: D5.10 (Tests 56–57).

Claim D5-08: Paper 2 compliance as a binary-testable prerequisite for Paper 3 participation

Statement:

Paper 2 compliance is a binary-testable prerequisite for full-aspect integration — the contributing party's Paper 2-compliant architecture can be verified as a pre-participation governance check.

Significance:

Paper 3 builds on Paper 2: a party participating in full-aspect integration must already instantiate the architectural commitments of Paper 2. This claim establishes that the prerequisite is not just a logical dependency but a verifiable governance check. A governance assessor can verify Paper 2 compliance before allowing participation, creating an enforceable entry condition. This is prior art for the pattern of prerequisite-as-governance-check, which is a reusable pattern across multi-paper architectural frameworks.

Established by: D5.10 (Test 60).

Claim D5-09: Health indicators and operational tests as complementary systems

Statement:

Health indicators and operational tests are complementary systems — health indicators provide continuous monitoring while operational tests provide periodic deep assessment, and their integration creates a unified governance quality assurance cycle.

Significance:

This claim articulates a structural relationship between two distinct governance tools. Health indicators are continuously updated signals; operational tests are periodic, deep-assessment instruments. The claim is that they are not alternatives but complements, and that their integration produces a governance quality assurance cycle more complete than either provides alone. This is prior art for the two-layer governance monitoring pattern.

Established by: D5.12 (health integration note).

Claim D5-10: Validation honesty as a governance-quality claim about the prior-art corpus

Statement:

The test library's three validation criteria — binary verifiability from records, distinct coverage across commitments, and non-trivial pass/fail conditions — combined with explicit acknowledgment of gaps, demonstrate intellectual honesty that strengthens the prior-art position.

Why this claim is distinctive:

Claims D5-01 through D5-09 make affirmative claims about what Paper 3's architecture is and how it can be tested. Claim D5-10 is a claim about the prior-art corpus itself. It asserts that the Phase D5 notes demonstrate intellectual honesty — by stating validation criteria explicitly, by applying those criteria consistently, and by acknowledging what the test library does not cover — and that this self-assessment property strengthens the corpus's credibility.

This matters for the defensive-publication purpose. A prior-art corpus assembled purely for coverage breadth, without explicit quality criteria and acknowledged limitations, is more vulnerable to challenge than one that documents its own assessment standards. Claim D5-10 is the claim that Phase D5 meets this higher standard.

Established by: D5.11 (validation criteria and acknowledged gaps).

4. Claims structure: meta-claim and instances

The ten claims above have a logical structure worth making explicit.

Claim D5-01 is the meta-claim. It asserts that binary testability from governance records is a property of Paper 3's governance commitments as a class. Claims D5-02, D5-03, D5-04, D5-07, and D5-08 are specific instances of this property, each demonstrating binary testability for a particular commitment or combination of commitments (speed-governance solution, N-ary completeness, competition-pair interaction, cross-paper isolation, and Paper 2 prerequisite, respectively).

Claims D5-05 and D5-06 extend beyond binary testability into the practical and quantitative dimensions: D5-05 asserts that governance quality is measurable as a continuous quantity, and D5-06 asserts that the test library is practically deployable across different contexts through scenario-specific subsets.

Claim D5-09 addresses the architectural relationship between two monitoring systems, positioning tests and health indicators as complementary rather than substitutable.

Claim D5-10 stands apart from the others: it is not a claim about Paper 3's architecture but about the prior-art corpus itself. Its function is to document the quality standard the Phase D5 notes were held to, which in turn supports the credibility of all ten claims.

5. Relationship to Phase D2 and Phase D3/D4 claims

Phase D2 established what the governance requirements of Paper 3's architecture are. Phase D5 claims do not restate those requirements; they establish that the requirements are testable. The relationship is additive: Phase D5 claims presuppose Phase D2's requirements as their subject matter and add testability as a new property of each.

Phase D3 established failure modes — the anti-patterns that arise when Paper 3’s commitments are violated. Phase D4 established composition pairs — the interactions between Paper 3’s commitments and commitments from Papers 1 and 2. Phase D5 Tests 56–60, which underlie Claims D5-07 and D5-08, explicitly test for cross-paper compliance and isolation. These tests connect the Phase D5 prior-art chain back to Phase D3 and D4 claims, providing a testability layer over the failure-mode and composition-pair coverage.

6. Forward pointer: D5.15

Note D5.15 is the final note in Phase D5. It provides: a summary of the 60-test library’s coverage across Paper 3’s six named claims; a consolidated review of the test subsets and their scenario applicability; a statement of Phase D5’s contribution to the overall Series D prior-art chain; and a transition to Phase D6 (boundary cases for Paper 3’s commitments).

This index (D5.14) is intended to be read alongside D5.15 as the two closing notes of Phase D5: the index for navigation, the closure note for synthesis.

7. How to cite Phase D5 claims

When citing a specific claim from Phase D5, the preferred form is:

Claim D5-0N, established in [note ID], in Li, W. (2026). *Phase D5 Key Prior-Art Claims Index*. May 15, 2026. ORCID: 0009-0004-8065-3235.

When citing the Phase D5 test library as a whole, the preferred form is:

Phase D5 60-test library, established in D5.07, in Li, W. (2026). Series D Phase D5 derivation notes. ORCID: 0009-0004-8065-3235.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

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